

A Polynomial-Time Approximation Scheme for Assortment Optimization under the Mallows Model

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Abstract

We study assortment optimization under a single Kendall–Mallows model with an arbitrarily positioned outside option and nonnegative product revenues. We develop a deterministic polynomial-time approximation scheme for the unconstrained problem, without bounds on the dispersion parameter or the ratio of positive revenues. The algorithm retains the first $O(1/\epsilon)$ offered alternatives in the central ranking and optimizes the remaining assortment by dynamic programming. A positive integral representation of choice probabilities shows that the retained alternatives control the relative error from compressing the remaining product factors. The compressed objective depends on the tail through its total and prefix weights; rounding these aggregates yields polynomially many states for each candidate head. This construction allows the tail to contain any number of products. The scheme also accommodates a cardinality constraint and one knapsack constraint, separately or jointly, while satisfying the original limits exactly. We derive bounds for pruning branches and report small-instance computational checks of the probability formulas, approximation steps, and optimization procedures.

Keywords: assortment optimization; Mallows model; approximation algorithms; dynamic programming.

1 Introduction

Assortment optimization determines which products a seller should offer when customers substitute among the available alternatives. In a ranking model, a customer draws a preference order over the products and an outside option, then selects the highest-ranked available alternative. Expected revenue therefore depends on the entire offered set. Even when the preference distribution has a compact description, the seller faces an optimization problem over exponentially many assortments.

The Kendall–Mallows model describes preferences through a central ranking and a dispersion parameter. Rankings closer to the central order receive higher probability. This structure has led to efficient procedures for evaluating assortment revenues and to approximation algorithms under several conditions. An unrestricted position of the outside option remains important for assortment design: products on either side of it can generate revenue, and the role of a product depends on both its rank and its revenue. A near-optimal assortment need not be identified by examining only an early segment of the central order.

We develop a polynomial-time approximation scheme (PTAS) for unconstrained assortment optimization under a single Mallows distribution. For any fixed $\epsilon \in (0, 1)$, the algorithm returns an assortment with expected revenue at least $(1 - \epsilon)$ times optimal in polynomial time. The outside option may occupy any position in the central ranking, the dispersion parameter may take any value in $[0, 1]$, and positive revenues may have any ratio. The result provides a polynomial-time approximation guarantee for the general single-model setting for which [Rieger and Segev \(2024\)](#) establish a quasi-polynomial-time scheme.

The main difficulty is to summarize the interaction among offered products without losing revenue from products with small choice probabilities. A uniform additive error in demand is insufficient when such products have large revenues. Our analysis instead controls the relative error of every product’s contribution. We express choice probabilities as positive integrals and retain the factors corresponding to the first K offered alternatives in the central order. These factors bound the relative error from replacing the remaining factors by an exponential. The error is $O(1/K)$ uniformly over the number and scale of the remaining factors.

This compression gives the algorithm its structure. For each candidate head, the contribution of a later offered product depends on the tail through a total weight and a prefix weight. After rounding these aggregates, we enumerate the total and choose the tail on a layered graph whose state records the prefix. A longest-path dynamic program solves each branch exactly for the approximate objective. Only the head has bounded size; the tail can include any number of products. Enumerating heads of size $O(1/\epsilon)$ and using relative integral approximations gives the PTAS.

The same graph supports assortment constraints. A cardinality limit adds a count coordinate to the state. A single knapsack constraint is handled by scaling path profits and retaining the minimum resource use for each profit label. The resulting scheme satisfies the cardinality and resource limits exactly, whether they are imposed separately or jointly. These extensions use the same approximation of choice probabilities as the unconstrained algorithm.

Head enumeration dominates the worst-case running time. We derive upper bounds on branch revenues that can be used for pruning, and we examine the construction through exact enumeration and high-precision calculations on small instances. These checks assess individual identities and algorithmic steps; they do not establish scalability on large product sets.

Section 2 discusses related work, and Section 3 presents the model and main result. Section 4 develops the approximation of assortment revenue. Section 5 gives the unconstrained PTAS, and Section 6 extends it to cardinality and knapsack constraints. Section 7 presents branch bounds and computational checks. Section 8 concludes. The appendices contain supporting proofs and evaluation procedures.

2 Related literature

The ranking distribution of [Mallows \(1957\)](#) provides the probabilistic basis for the model. [Désir et al. \(2021\)](#) study Mallows-smoothed choice models, including mixtures, and develop choice-probability evaluation procedures, a mixed-integer formulation, and approximation schemes. Their PTAS assumes that the outside option is last in each central ranking. Under that assumption, the method also accommodates downward-closed assortment constraints, including cardinality, knapsack, and matroid constraints.

[Rieger and Segev \(2024\)](#) establish a quasi-polynomial-time approximation scheme for the general single-Mallows assortment problem, allowing an arbitrary outside-option position and unrestricted dispersion and revenue ratios. This is the closest benchmark for our unconstrained problem. Theorem 3.1 gives a PTAS for that setting. Its structural ingredient is a uniform relative approximation obtained by retaining a short head and compressing the tail. The algorithm then optimizes over tails of unrestricted size through rounded aggregate weights.

The constrained extension in Section 6 applies to a single Mallows distribution with an arbitrary outside-option position, a cardinality limit, and one knapsack constraint. The outside-last results of [Désir et al. \(2021\)](#) cover broader constraint families and mixtures under their ranking assumption. Our extension relies on the single central order and a scalar resource label; it does not cover arbitrary downward-closed families or mixtures.

For the knapsack extension, we use profit scaling in an acyclic path problem. This approach is related to the resource-constrained path approximation schemes of [Hassin \(1992\)](#). We provide a self-contained maximization argument for the graph generated by the assortment problem.

The Mallows analysis supplies the graph and its approximation of revenue; the path procedure enforces the resource budget.

3 Model and main result

There are N alternatives, indexed by their positions in a fixed central ranking $1, \dots, N$. Alternative $o \in [N]$ is the outside option. It is always available and has revenue $r_o = 0$; every product has revenue $r_i \geq 0$. For $0 < q < 1$, a permutation σ of the entire universe has probability

$$\Pr(\sigma) = \frac{q^{\text{inv}(\sigma)}}{Z_N(q)}, \quad Z_n(q) = \prod_{j=1}^n (1 + q + \dots + q^{j-1}), \quad Z_0(q) = 1. \quad (3.1)$$

Here $\text{inv}(\sigma)$ is its inversion count relative to the central ranking. At $q = 0$, the ranking is the central order deterministically; at $q = 1$, it is uniform over all permutations. The distribution and product revenues are known when the seller chooses the assortment. All numerical input data are rational numbers encoded in binary.

We include the outside option in the offer set $A \subseteq [N]$. A customer chooses the first alternative in A encountered in σ . Let $P_i(A)$ be the probability of choosing $i \in A$. The expected revenue and the unconstrained optimization problem are

$$R(A) = \sum_{i \in A} r_i P_i(A), \quad \text{OPT} = \max_{A \subseteq [N]: o \in A} R(A). \quad (3.2)$$

The choice probabilities are induced by rankings of the full universe; in particular, the original rank gaps between offered alternatives are retained. Restricting the offer set does not replace the preference distribution by a Mallows distribution on that smaller set.

Theorem 3.1 (PTAS for unconstrained assortment optimization). *For every $\epsilon \in (0, 1)$, there is a deterministic algorithm that returns an assortment $\hat{A} \ni o$ satisfying*

$$R(\hat{A}) \geq (1 - \epsilon) \text{OPT}.$$

For each fixed ϵ , the algorithm runs in time polynomial in the input size. It allows any position of the outside option and any $q \in [0, 1]$, without an upper bound on the ratio of positive revenues.

The algorithm enumerates heads of size $K = \lceil 1/\epsilon \rceil + 4$ and solves a polynomial-size dynamic program for each head and rounded total. Its combinatorial work is $O(N^{K+5})$, with polynomial time for score evaluation. Thus the exponent depends on the desired accuracy, as permitted by a PTAS. Section 6 shows that the same approximation guarantee holds under a cardinality limit and one knapsack constraint. The demand model throughout consists of a single Mallows component.

4 Structural approximation of assortment revenue

The approximation has two stages. We first replace the factors associated with later offered alternatives by an exponential while retaining a short head. We then round only the aggregate weights in that exponential. Both operations produce pointwise lower bounds on revenue with relative losses that depend on the retained head size.

4.1 A positive representation of choice probabilities

For $0 < q < 1$, set

$$w_i = q^{i-1}, \quad c_q = \frac{1-q}{-\log q}, \quad D_n(t) = \prod_{j=0}^n (1 + q^j t). \quad (4.1)$$

The following representation separates the original rank weights from the assortment-dependent factors. Its positivity will permit relative error control for each revenue contribution.

Proposition 4.1 (Positive choice integral). *For every nonempty $A \subseteq [N]$ and $i \in A$,*

$$P_i(A) = c_q w_i \int_0^\infty \prod_{\substack{j \in A \\ j \leq i}} (1 + w_j t)^{-1} \prod_{\substack{j \in A \\ j \geq i}} (1 + q w_j t)^{-1} dt. \quad (4.2)$$

In particular, i contributes two factors, with coefficients w_i and $q w_i$.

The proof is given in Appendix A.

For $q = 1$, all full rankings are equally likely, so $P_i(A) = 1/|A|$. Formula (4.2) continues to hold with $c_q = w_i = 1$, since its integrand then equals $(1 + t)^{-|A|-1}$. All subsequent arguments apply at $q = 1$ with these definitions. The case $q = 0$ is treated in Section 5.5.

The weights w_i retain the positions in the original central order. Thus the representation applies directly to nonconsecutive offer sets. Because the integrand is nonnegative, a common relative lower bound on the integrals remains a relative lower bound after multiplication by revenues and summation.

4.2 Uniform relative error from compression

The next lemma explains why retaining a short head is sufficient. The retained factors concentrate the normalized integrand near the origin, where the product-to-exponential approximation has a controlled error. The bound depends on the number of retained factors and is uniform over the number and size range of the factors being compressed.

Lemma 4.2 (Product-to-exponential compression). *Let $K > 3$ be an integer. Let $C(t) = \prod_\ell (1 + a_\ell t)^{-1}$ have at least K coefficients $a_\ell \geq a > 0$; any additional coefficients are nonnegative. Let $0 \leq b_j \leq a$ and $B_0 = \sum_j b_j$. Define*

$$I = \int_0^\infty C(t) \prod_j (1 + b_j t)^{-1} dt, \quad J = \int_0^\infty C(t) e^{-B_0 t} dt.$$

Then, with $\alpha_K = 1/[4(K - 3)]$,

$$(1 - \alpha_K)I \leq J \leq I. \quad (4.3)$$

Proof. Let f denote the integrand of I , and put $z = B_0/a$. Since $x/(1 + xt)$ increases in x and $b_j \leq a$,

$$-\frac{f'(t)}{f(t)} \geq \frac{Ka}{1 + at} + \sum_j \frac{b_j}{1 + at} = \frac{a(K + z)}{1 + at}.$$

Thus $f(t)/(1 + at)^{-(K+z)}$ is nonincreasing. After normalization, this monotone likelihood ratio implies that f/I is stochastically no larger than the normalized density proportional to $(1 + at)^{-(K+z)}$: the two densities have at most one crossing, from above to below. In particular,

$$\mathbb{E}_f[t^2] \leq \frac{2}{a^2(K + z - 2)(K + z - 3)}. \quad (4.4)$$

Here and below the subscript f means normalization by its integral. For $x \geq 0$, $0 \leq x - \log(1 + x) \leq x^2/2$. Therefore

$$0 \leq D(t) := B_0 t - \sum_j \log(1 + b_j t) \leq \frac{t^2}{2} \sum_j b_j^2 \leq \frac{a B_0 t^2}{2}.$$

Since $J/I = \mathbb{E}_f[e^{-D(t)}]$,

$$0 \leq 1 - J/I \leq \frac{aB_0}{2} \mathbb{E}_f[t^2] \leq \frac{z}{(K+z-2)(K+z-3)} \leq \frac{1}{4(K-3)}.$$

The last inequality follows from $z/(c+z)^2 \leq 1/(4c)$ for $c = K-3 > 0$. $\square$

Lemma 4.3 (Additive shift). *Suppose C has at least $K > 2$ factors with coefficients at least $a_0 > 0$, and any additional nonnegative factors. For $L(s) = \int_0^\infty e^{-st} C(t) dt$ and $s, d \geq 0$,*

$$\left(1 - \frac{d}{a_0(K-2)}\right) L(s) \leq L(s+d) \leq L(s). \quad (4.5)$$

The proof is given in Appendix B.

4.3 Retaining the first offered alternatives

We now apply Lemma 4.2 to the assortment problem. Fix

$$K = \lceil 1/\epsilon \rceil + 4. \quad (4.6)$$

Enumerate every assortment containing o with fewer than K alternatives separately and evaluate its true revenue exactly using Appendix A.3. Every other assortment has a unique head H consisting of its first K offered alternatives. Write $h = \max H$ and $A = H \cup T$, where $T \subseteq \{h+1, \dots, N\}$. We enumerate all possible K -element heads and enforce

$$o \leq h \implies o \in H, \quad o > h \implies o \in T. \quad (4.7)$$

No unselected position at or before h is later inserted into T .

For a fixed head define

$$L_H(s) = \int_0^\infty e^{-st} \prod_{j \in H} (1 + w_j t)^{-1} dt, \quad (4.8)$$

$$L_{H,i}(s) = \int_0^\infty e^{-st} \prod_{\substack{j \in H \\ j \leq i}} (1 + w_j t)^{-1} \prod_{\substack{j \in H \\ j \geq i}} (1 + q w_j t)^{-1} dt, \quad i \in H. \quad (4.9)$$

Let

$$W(T) = \sum_{j \in T} w_j, \quad U_i(T) = \sum_{\substack{j \in T \\ j < i}} w_j.$$

The compressed revenue is

$$\tilde{R}_H(T) = c_q \left[\sum_{i \in H} r_i w_i L_{H,i}(qW(T)) + \sum_{i \in T} r_i w_i L_H(qW(T) + (1-q)U_i(T) + w_i) \right]. \quad (4.10)$$

Proposition 4.4 (Pointwise compressed approximation). *For every tail T ,*

$$(1 - \alpha_K) R(H \cup T) \leq \tilde{R}_H(T) \leq R(H \cup T). \quad (4.11)$$

The proof is given in Appendix B.

For a fixed head, the approximate revenue of a tail product depends on the tail only through $W(T)$ and $U_i(T)$. Conditioning on the total $W(T)$ will therefore leave a sequential decision problem whose state is a prefix weight. The next step makes both aggregates discrete without introducing a dependence on the ratio of the largest and smallest rank weights.

4.4 Discretizing aggregate weights

Put $a = w_h$ and use the rational grid

$$\Delta = \frac{a}{4N}, \quad z_i = \left\lceil \frac{w_i}{\Delta} \right\rceil, \quad \bar{w}_i = \Delta z_i \quad (i > h), \quad M = 4N^2. \quad (4.12)$$

Because $0 < w_i \leq a$, we have $1 \leq z_i \leq 4N$ and $\sum_{i \in T} z_i \leq M$. Replace only W and U_i in (4.10) by $\bar{W} = \sum_{j \in T} \bar{w}_j$ and $\bar{U}_i = \sum_{j \in T, j < i} \bar{w}_j$. Keep the multipliers $r_i w_i$ and the standalone w_i in each exponent unchanged. Call the result $\bar{R}_H(T)$.

Proposition 4.5 (Weight rounding). *With $\beta_K = 1/[4(K-2)]$,*

$$(1 - \beta_K) \tilde{R}_H(T) \leq \bar{R}_H(T) \leq \tilde{R}_H(T). \quad (4.13)$$

The proof is given in Appendix B.

5 The unconstrained approximation algorithm

For a fixed head and rounded total weight, the approximate objective is additive along a layered graph. We first describe how its coefficients are evaluated, then give the dynamic program and prove Theorem 3.1.

5.1 Evaluating the approximate revenue

The integral coefficients are computed from below with a relative error $\delta = \epsilon/4$. Relative accuracy preserves the contribution of a product even when its choice probability is small and its revenue is large.

Lemma 5.1 (Relative lower approximation). *Let $m \geq 2$, $c_1, \dots, c_m > 0$, $s \geq 0$, and $0 < \delta < 1$ be rational. For*

$$L = \int_0^\infty f(t) dt, \quad f(t) = e^{-st} \prod_{\ell=1}^m (1 + c_\ell t)^{-1}, \quad (5.1)$$

one can compute a rational $\underline{L}$ satisfying

$$(1 - \delta)L \leq \underline{L} \leq L \quad (5.2)$$

in time polynomial in the input size and $1/\delta$.

The construction and proof are given in Appendix D.

Apply the lemma to every integral with a positive revenue multiplier. Zero multipliers remain zero. Define $F(A) = R(A)/c_q$, and let $\underline{F}_H(T)$ be the rounded, compressed score with these lower integral values and with the common factor c_q omitted. Propositions 4.4 and 4.5 give

$$\rho F(H \cup T) \leq \underline{F}_H(T) \leq F(H \cup T), \quad \rho = (1 - \delta)(1 - \beta_K)(1 - \alpha_K). \quad (5.3)$$

The same approximation factor applies to every tail and every head. We use $\underline{F}_H$ to optimize within a branch and compare the resulting assortments by their true revenues. Appendix A.3 gives an exact revenue evaluation procedure using $O(N^3)$ arithmetic operations per assortment. Thus the comparison among returned candidates introduces no additional approximation loss.

5.2 A layered graph for a fixed head and total

Fix a head H and enumerate $Q \in \{0, \dots, M\}$, where $M = 4N^2$. The branch consists of tails satisfying $\sum_{i \in T} z_i = Q$. Their head contribution is the constant

$$b_H(Q) = \sum_{i \in H} r_i w_i \underline{L}_H(q\Delta Q). \quad (5.4)$$

Scan positions $i = h + 1, \dots, N$ in order. A state records the cumulative rounded weight p of the selected alternatives before i . Selecting i has score

$$b_i(p; Q) = r_i w_i \underline{L}_H(q\Delta Q + (1 - q)\Delta p + w_i) \quad (5.5)$$

and sends p to $p + z_i$. Skipping i keeps p unchanged and has zero score. Omit selection edges with $p + z_i > Q$. If $i = o$, prohibit the skip edge; the outside option has zero score but contributes its rank weight to the state. The source has $p = 0$, and the terminal state has $p = Q$ after all positions have been processed. Discard branches with no source–terminal path.

Every path selects a unique tail, and every tail in the branch defines a path. Its total edge score is $\underline{F}_H(T) - b_H(Q)$. Future edge scores depend on previous choices only through p , so no other state coordinate is needed in the unconstrained problem.

Proposition 5.2 (Exact optimization within a branch). *For each H, Q , a longest-path dynamic program on the layered graph maximizes $\underline{F}_H(T)$ over all tails in the branch.*

Proof. Every transition advances one layer, so the graph is acyclic. In topological order, apply

$$V(v) = \max_{e=(u,v)} \{V(u) + b_e\},$$

with $V(\text{source}) = 0$ and $-\infty$ for unreachable states. The path–tail correspondence shows that the terminal value is the maximum tail score. Adding the constant $b_H(Q)$ gives the maximum of $\underline{F}_H$. Predecessor pointers recover a maximizing tail. $\square$

5.3 Algorithm and approximation guarantee

Every assortment of size at least K has a unique head and rounded total. Algorithm 1 examines that branch and optimizes its approximate revenue; smaller assortments are evaluated directly.

Algorithm 1 (Unconstrained assortment PTAS for $0 < q \leq 1$).

1. Set $K = \lceil 1/\epsilon \rceil + 4$ and $\delta = \epsilon/4$. Initialize the best candidate to $\{o\}$, with revenue zero.
2. Enumerate all $A \ni o$ with $|A| < K$, evaluate their true revenues using Appendix A.3, and update the best candidate.
3. Enumerate all K -element heads H consistent with (4.7). Form the grid in (4.12). For each $Q = 0, \dots, M$, construct the graph in Section 5.2.
4. For each branch with a source–terminal path, find a longest path and its tail T . Evaluate $R(H \cup T)$ and update the best candidate.
5. Return the candidate with largest true revenue.

If $K > N$, Step 2 covers every assortment and Step 3 is empty.

Approximation guarantee in Theorem 3.1 for $0 < q \leq 1$. Let A^* be an optimal assortment. If $|A^*| < K$, it is evaluated exactly. Otherwise, let H^* be its first K offered alternatives, $T^* =$

$A^* \setminus H^*$, and $Q^* = \sum_{i \in T^*} z_i$ for the grid of H^* . This branch is enumerated and contains the path corresponding to T^* . By Proposition 5.2, the returned tail $\hat{T}$ satisfies

$$F(H^* \cup \hat{T}) \geq \underline{F}_{H^*}(\hat{T}) \geq \underline{F}_{H^*}(T^*) \geq \rho F(A^*).$$

The final comparison ensures $R(\hat{A}) \geq R(H^* \cup \hat{T})$. Multiplying by $c_q > 0$ gives

$$\begin{aligned} R(\hat{A}) &\geq (1 - \delta)(1 - \beta_K)(1 - \alpha_K) \text{OPT} \\ &\geq (1 - \epsilon/4)^3 \text{OPT} \geq (1 - \epsilon) \text{OPT}. \end{aligned}$$

Here $\alpha_K, \beta_K \leq \epsilon/4$, and $\prod_j (1 - x_j) \geq 1 - \sum_j x_j$ for $x_j \in [0, 1]$. The construction includes the outside option in every returned assortment. The proof also covers $\text{OPT} = 0$, since it never divides by optimal revenue. $\square$

5.4 Running time

There are at most N^K heads and $M + 1 = O(N^2)$ rounded totals per head. For a fixed head and total, the graph has $O(NM) = O(N^3)$ states and edges. The longest-path computation therefore uses $O(N^{K+5})$ edge updates across all branches. Integral values are evaluated once per edge and reused. By Lemma 5.1, each evaluation takes polynomial time in the input size and $1/\epsilon$.

There are at most $O(N^{K+2})$ branch candidates, each requiring $O(N^3)$ arithmetic operations to evaluate its true revenue. The smaller-assortment enumeration also fits within this bound. Since $K = O(1/\epsilon)$, the overall algorithm is polynomial for each fixed ϵ . Appendix D.2 provides the arithmetic details. Only reachable rounded totals need be examined; a subset-sum dynamic program on $\{0, \dots, M\}$, with mandatory outside-option inclusion, identifies them in polynomial time.

5.5 Boundary values of the dispersion parameter

For $q = 0$, the full ranking equals the central ranking. A purchased product must precede o . Offering only that product together with o preserves the purchase, so an optimal assortment consists of the highest-revenue product with index $i < o$, together with o . Return $\{o\}$ if no such product has positive revenue. This exact procedure completes Theorem 3.1.

At $q = 1$, every offered alternative is chosen with probability $1/|A|$. For a fixed number k of offered products, the best assortment contains the k highest-revenue products and earns their revenue sum divided by $k + 1$. Checking $k = 0, \dots, N - 1$ gives an exact algorithm. The PTAS construction is also valid at $q = 1$, using $c_q = w_i = 1$.

6 Cardinality and knapsack constraints

Suppose that product i consumes a nonnegative amount d_i of a static resource, such as shelf space, and let $d_o = 0$. We consider

$$|A \setminus \{o\}| \leq C, \quad \sum_{i \in A} d_i \leq B, \tag{6.1}$$

where $C \geq 0$ is an integer and $B \geq 0$. Either constraint may be omitted. Let $\mathcal{F}$ be the family of assortments containing o and satisfying the imposed constraints, and write $\text{OPT}_{\mathcal{F}} = \max_{A \in \mathcal{F}} R(A)$. The revenue approximations in Section 4 hold for every assortment and hence for every member of $\mathcal{F}$.

Theorem 6.1 (PTAS with assortment constraints). *For every $\epsilon \in (0, 1)$, a deterministic algorithm returns $\hat{A} \in \mathcal{F}$ with*

$$R(\hat{A}) \geq (1 - \epsilon) \text{OPT}_{\mathcal{F}}$$

in polynomial time for fixed ϵ . The returned assortment satisfies the original cardinality and resource limits without relaxation.

6.1 A cardinality constraint

Restrict the smaller-assortment enumeration to feasible assortments and discard heads containing more than C products. For a retained head, let $C_H = C - |H \setminus \{o\}|$. Add a coordinate $k \in \{0, \dots, \min(C_H, N)\}$ that counts selected tail products. Selecting a product increases k by one; selecting o or skipping a product leaves it unchanged. Prohibit transitions exceeding C_H .

Every terminal state with $p = Q$ is accepted, regardless of its count. A zero-score edge from each accepted terminal to a common sink gives one source–sink problem. The graph remains acyclic, and the longest-path recurrence exactly maximizes the branch score subject to the cardinality limit. Thus cardinality introduces no additional approximation loss.

6.2 A knapsack constraint

Discard heads with $\sum_{i \in H} d_i > B$. For each retained head, let $B_H = B - \sum_{i \in H} d_i$ be the remaining budget. Give the selection edge for i resource use d_i , and give skip and sink edges resource use zero. If a cardinality limit is also imposed, use the count coordinate above. Otherwise, retain just the prefix-weight state. Every source–sink path has at most N edges, including an optional final sink edge.

Lemma 6.2 (Resource-constrained path FPTAS). *Let an acyclic source–sink graph have nonnegative edge profits b_e and resource uses d_e , with at most N edges on any source–sink path. Given $B_H \geq 0$ and $0 < \eta < 1$, an algorithm finds a feasible path of profit at least $(1 - \eta)G^*$, or determines that none exists. Here G^* is the maximum feasible path profit. The algorithm performs $O(|E|N^2/\eta)$ label updates, in addition to acyclic shortest-path computations.*

The proof is given in Appendix C.

The procedure first removes edges that belong to no feasible path. If P is the largest remaining edge profit, then $P \leq G^* \leq NP$ because profits are nonnegative and some feasible path contains that edge. Rounding profits down in units of $\eta P/N$ produces $O(N^2/\eta)$ path-profit labels. For each vertex and label, the dynamic program stores the minimum resource use. Resource amounts are kept exact, so the number of labels does not depend on the numerical magnitude of B_H , and the budget is never relaxed.

Apply Lemma 6.2 with $\eta = \epsilon/4$. Since the head contribution is nonnegative, adding it preserves the guarantee:

$$b_H(Q) + (1 - \eta)G^* \geq (1 - \eta)(b_H(Q) + G^*).$$

For each branch, evaluate the true revenue of the returned assortment. Choose the largest-revenue candidate among these and all feasible smaller assortments, as in Algorithm 1.

6.3 Approximation guarantee and running time

Proof of Theorem 6.1. For $0 < q \leq 1$, let A^* maximize revenue over $\mathcal{F}$. If $|A^*| < K$, it is evaluated directly. Otherwise its head H^* satisfies every head restriction, and its tail T^* defines a feasible path in the branch with total Q^* . The count coordinate, if present, retains this path. The branch algorithm returns a feasible $\hat{T}$ with

$$\underline{E}_{H^*}(\hat{T}) \geq (1 - \eta)\underline{E}_{H^*}(T^*),$$

Table 1: Upper bounds on state and label updates, where $K = \lceil 1/\epsilon \rceil + 4$.

Assortment constraints	Total updates across heads and totals
None	$O(N^{K+5})$
Cardinality	$O(N^{K+6})$
One knapsack	$O(N^{K+7}/\epsilon)$
Cardinality and one knapsack	$O(N^{K+8}/\epsilon)$

where $\eta = \epsilon/4$ when a knapsack constraint is imposed and $\eta = 0$ otherwise. By (5.3) and final revenue comparison,

$$R(\hat{A}) \geq (1 - \eta)\rho \text{OPT}_{\mathcal{F}} \geq (1 - \epsilon/4)^4 \text{OPT}_{\mathcal{F}} \geq (1 - \epsilon) \text{OPT}_{\mathcal{F}}.$$

All returned paths satisfy the imposed constraints exactly.

For $q = 0$, removing every product except the purchased one preserves both revenue and feasibility. It therefore suffices to check $\{o\}$ and all feasible sets $\{i, o\}$ with $i < o$. This also covers zero budgets, zero costs, and $C = 0$. At $q = 1$, the general construction remains valid. If only a cardinality limit is present, the exact algorithm of Section 5.5 can instead check $0 \leq k \leq \min\{C, N - 1\}$.

For each H, Q , the count coordinate increases the graph size by at most N . The knapsack procedure uses $O(N^2/\epsilon)$ labels per vertex. Table 1 gives the resulting numbers of updates, with polynomial time for coefficient evaluation and each update. These bounds are polynomial for fixed ϵ , proving the theorem. $\square$

7 Implementation and computational verification

7.1 Branch bounds and the choice of head size

The enumeration of heads is the largest combinatorial cost. For a fixed head and rounded total, let $U_{H,Q}$ be the maximum lower score on the layered graph, including $b_H(Q)$. In the unconstrained problem, it is obtained by the same longest-path dynamic program used to select the branch candidate. Equation (5.3) implies

$$R(H \cup T) \leq c_q U_{H,Q} / \rho$$

for every completion in that branch. A maximized lower approximation of revenue therefore also provides an upper bound on the true branch optimum after adjustment by its approximation factor.

For the constrained problem, compute $U_{H,Q}$ after relaxing the knapsack constraint, while retaining the cardinality and outside-option restrictions. The same upper bound applies to every feasible completion. Lagrangian penalties for resource use can tighten it. Appendix E gives these bounds and their evaluation. A branch can be discarded if its upper bound does not exceed the revenue of the best assortment already found. The bounds preserve the approximation guarantees, but no reduction in worst-case running time is asserted.

The prescribed head size can be substantial: $\epsilon = 0.1$ gives $K = 14$. Using a smaller head reduces enumeration and yields the explicit approximation factor in Appendix E, with a corresponding loss in the guarantee. This tradeoff motivates implementation choices, but its computational value must be assessed on instances larger and more varied than those below.

Table 2: Complete-algorithm checks on eight alternatives. Returned revenues and optimal revenues coincide in these six instances.

q	Outside rank o	C	B	$ \hat{A} $	$R(\hat{A}) = \text{OPT}_{\mathcal{F}}$	Branches
1	1	–	–	8	7/8	28
1	5	6	–	7	6/7	27
1	8	–	6	7	6/7	28
1	5	6	6	7	6/7	27
1/2	1	–	6	7	63/127	28
1/2	8	6	6	7	10752/10795	28

7.2 Design of the computational checks

The computations examine distinct parts of the construction using independent finite references. Complete permutation enumeration supplies exact choice probabilities. Enumeration of feasible paths checks the dynamic programs, and enumeration of feasible assortments supplies true optimal revenues. The probability comparisons use rational arithmetic. The numerical integral comparisons use 50 decimal digits and a relative tolerance of 10^{-35} .

The complete-algorithm examples have $N = 8$ alternatives, unit product revenues and unit resource uses, and zero revenue and resource use for the outside option. We set $\epsilon = 3/4$, so $K = 6$, and vary the outside position, dispersion, and constraint combination as shown in Table 2. An omitted constraint is denoted by a dash. These small examples permit exhaustive comparison and include solutions from the large-assortment branches. Their deliberately loose accuracy parameter makes those branches observable at this instance size.

In these six cases, branch profits are obtained from high-precision quadrature and converted to 35-digit rational values. Resource comparisons, profit labels, and final true-revenue comparisons use exact rational arithmetic. The examples therefore test the complete combinatorial construction, but they are not full implementations of the certified integral oracle on every edge. The certified oracle is exercised separately on eight integrals and on every score in two additional branch-bound examples.

7.3 Results and interpretation

All six complete-algorithm examples attain their exactly enumerated optimum. The checks of 3,076 choice probabilities and 49 symbolic identities also agree exactly with their references. Across 204 numerical integral identities, the largest relative error is below $6 \cdot 10^{-51}$. The separate path tests include a constructed profit-rounding tie that returns 19/10 against an optimum of 2. Its relative loss of 1/20 confirms that profit scaling can lose revenue even when the approximation guarantee holds.

For the certified branch-bound checks, we use a head of six alternatives and two tail products, with $q \in \{1, 99/100\}$. Five nonnegative rational Lagrangian multipliers produce ten bounds, all verified against exact branch optima using rational arithmetic. The relaxed-capacity and penalized bounds are not necessarily tight. For example, when $q = 1$, the branch optimum is 8/7, the zero-multiplier bound is approximately 1.84610, and the bound at $\lambda = 1/2$ is approximately 1.64343. This example checks validity and illustrates possible tightening; it does not measure overall pruning speed.

These results support consistency of the formulas and implementations on the specified finite cases. The approximation theorem follows from the analysis in Sections 4–6. Neither the unit-revenue examples nor the component checks establish practical performance for large, heterogeneous instances.

8 Conclusion

We have developed a PTAS for unconstrained assortment optimization under a single Mallows model with an arbitrary outside-option position, dispersion parameter, and ratio of positive revenues. The main structural result is a uniform relative approximation of assortment revenue after the first $O(1/\epsilon)$ offered alternatives are retained. The remaining assortment can be represented through total and prefix weights and optimized by a longest-path dynamic program. The tail can be large, and its products retain their positions in the original central ranking.

Cardinality and one knapsack constraint can be incorporated into the same graph. A product-count coordinate handles cardinality exactly, while profit scaling and minimum-resource labels give the knapsack extension. Both extensions preserve the original feasibility limits and the PTAS guarantee.

The enumeration of heads limits the practical scope of the construction. The branch bounds permit safe pruning, but the small-instance checks do not establish its effectiveness at scale. Reducing the enumeration cost while preserving relative control of revenue is a natural algorithmic question. Mixtures of Mallows distributions and multiple resource budgets require additional ideas: the present dynamic program relies on a single central order and, for capacity, a scalar minimum-resource label.

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A Choice probabilities and exact revenue evaluation

A.1 Geometric-product moments

The probability representation rests on the following moment identity. For $0 < q < 1$, use the notation in (4.1).

Lemma A.1 (Geometric-product moments). *For $n \geq 1$ and $0 \leq k \leq n-1$,*

$$c_q \int_0^\infty \frac{t^k}{D_n(t)} dt = q^{-k(k+1)/2} \frac{Z_k(q) Z_{n-k-1}(q)}{Z_n(q)}. \quad (\text{A.1})$$

Proof. Write $F_{n,k}$ for the integral on the left before multiplication by c_q . For $g(t) = D_{n-1}(t)^{-1}$,

$$\frac{g(qt) - g(t)}{t} = \frac{1 - q^n}{D_n(t)}.$$

Integrating on $[a, b]$ and changing variables in the first term gives

$$\int_a^b \frac{g(qt) - g(t)}{t} dt = \int_{qa}^a \frac{g(u)}{u} du - \int_{qb}^b \frac{g(u)}{u} du.$$

Since $g(0) = 1$ and $g(\infty) = 0$, the limit as $a \downarrow 0$, $b \uparrow \infty$ is $-\log q$. Thus $F_{n,0} = (-\log q)/(1 - q^n)$. For $1 \leq k \leq n-1$, both of the following integrals converge:

$$F_{n,k-1} + q^n F_{n,k} = F_{n-1,k-1}, \quad F_{n,k-1} + F_{n,k} = q^{-k} F_{n-1,k-1}.$$

Their difference yields

$$F_{n,k} = \frac{q^{-k} - 1}{1 - q^n} F_{n-1,k-1}.$$

Starting from $F_{n-k,0}$ and substituting $Z_j = (1 - q)^{-j} \prod_{\ell=1}^j (1 - q^\ell)$ proves (A.1). $\square$

A.2 Proof of the positive integral representation

Proof of Proposition 4.1. Let $U = [N] \setminus A$, and fix the set $B \subseteq U$ of alternatives appearing before i in a full ranking in which i wins. Put $k = |B|$. The ranking consists of the elements of B , then i , then its remaining elements. Summing the weights of the two internal permutations gives $Z_k Z_{N-k-1}$. The inversions between the three parts number

$$\begin{aligned} E(B, i) &= \sum_{b \in B} (b-1) - \frac{k(k-1)}{2} + (i-1) - |\{b \in B : b < i\}| \\ &= i-1 + \sum_{b \in B} (b-1 + \mathbf{1}_{\{b > i\}}) - \frac{k(k+1)}{2}. \end{aligned}$$

Consequently,

$$P_i(A) = \frac{w_i}{Z_N} \sum_{B \subseteq U} q^{-k(k+1)/2} Z_k Z_{N-k-1} \prod_{b \in B} q^{b-1 + \mathbf{1}_{\{b > i\}}}. \quad (\text{A.2})$$

For the full offer set, the denominator in (4.2) is $D_N(t)$. Removing an unoffered $b < i$ cancels $1 + q^{b-1}t$, and removing $b > i$ cancels $1 + q^b t$. Hence its integrand is

$$\frac{\prod_{b \in U} (1 + q^{b-1 + \mathbf{1}_{\{b > i\}}} t)}{D_N(t)}.$$

Expand the numerator and apply Lemma A.1 term by term. The result is exactly (A.2). $\square$

A.3 Exact rational revenue evaluation

For $0 < q < 1$, let $(q; q)_j = \prod_{\ell=1}^j (1 - q^\ell)$ with $(q; q)_0 = 1$. For a fixed offered $i \in A$, expand

$$\prod_{\substack{j \notin A \\ j < i}} (1 + q^{j-1} z) \prod_{\substack{j \notin A \\ j > i}} (1 + q^j z) = \sum_{k=0}^{N-|A|} e_k^{(i)} z^k. \quad (\text{A.3})$$

Proposition 4.1 and Lemma A.1 imply the exact formula

$$\begin{aligned} P_i(A) &= \frac{w_i}{Z_N} \sum_{k=0}^{N-|A|} e_k^{(i)} q^{-k(k+1)/2} Z_k Z_{N-k-1} \\ &= (1-q) w_i \sum_{k=0}^{N-|A|} e_k^{(i)} q^{-k(k+1)/2} \frac{(q; q)_k (q; q)_{N-k-1}}{(q; q)_N}. \end{aligned} \quad (\text{A.4})$$

In contrast to its integral representation, this expression is rational for rational q and contains no logarithm.

The coefficients in (A.3) can be computed by a product-of-linear-factors recurrence. Initialize $e_0 = 1$ and all other coefficients to zero. For each factor $1 + vz$, update in descending degree order by

$$e_k \leftarrow e_k + v e_{k-1}.$$

There are at most N factors and at most $N + 1$ coefficients, so this requires $O(N^2)$ rational operations for one offered alternative. Precompute Z_j and the required powers of q ; summing (A.4) costs $O(N)$ more operations. Repeating for every $i \in A$ and summing $r_i P_i(A)$ computes $R(A)$ in $O(N^3)$ rational operations.

Let $\mathcal{L}$ denote the total binary encoding length of the numerical input data, including the requested accuracy. The bit lengths in this evaluation are polynomial in N and $\mathcal{L}$. If $q = a/b$ is in lowest terms, all exponents in these products and coefficients have magnitude $O(N^2)$. The elementary symmetric coefficients are sums of at most 2^N monomials, and the partition functions have degree $O(N^2)$ and total coefficient mass at most $N!$. Common-denominator representations and products of polynomially many such rational quantities therefore have polynomial bit length in $N, \mathcal{L}$. Exact signs and comparisons of candidate revenues are thus polynomial-time operations. At $q = 1$, use $P_i(A) = 1/|A|$; at $q = 0$, use the deterministic rule from Section 5.5.

Let $\mathcal{C}$ be the candidates generated by the branch algorithms together with the small assortments. Exact final selection satisfies

$$R(\arg \max_{A \in \mathcal{C}} R(A)) \geq R(A') \quad \text{for every } A' \in \mathcal{C}.$$

In particular it weakly dominates selecting the same candidates by their lower scores, while preserving the overall PTAS complexity.

B Supporting structural proofs

B.1 Additive shifts

Proof of Lemma 4.3. The logarithmic derivative of $e^{-st} C(t)$ is at most $-K a_0 / (1 + a_0 t)$. The likelihood-ratio argument from the proof of Lemma 4.2, now with reference density proportional to $(1 + a_0 t)^{-K}$, gives $\mathbb{E}_s[t] \leq 1/[a_0(K - 2)]$. Use $L(s + d)/L(s) = \mathbb{E}_s[e^{-dt}] \geq 1 - d \mathbb{E}_s[t]$. Monotonicity gives the upper bound. $\square$

B.2 Head and tail compression

Proof of Proposition 4.4. For $i \in H$, retain the $K + 1$ factors in (4.9). Their coefficients are at least qw_h , whereas all removed factors have coefficients $qw_j \leq qw_h$. Their sum is $qW(T)$, so Lemma 4.2 applies. For $i \in T$, retain the K head factors $w_j \geq w_h$. All remaining coefficients are at most w_h , and their sum equals

$$\sum_{\substack{j \in T \\ j < i}} w_j + w_i + q \sum_{\substack{j \in T \\ j \geq i}} w_j = qW(T) + (1 - q)U_i(T) + w_i.$$

Apply the same lemma to each choice integral, then sum using $r_i w_i \geq 0$. $\square$

B.3 Rounding aggregate weights

Proof of Proposition 4.5. Every rounded sum increases by at most $N\Delta = a/4$. For a head term its exponent increases by at most $qa/4$; its core has at least K coefficients at least qa . Lemma 4.3 gives loss at most $1/[4(K - 2)]$. For a tail term the shift is at most $q(a/4) + (1 - q)(a/4) = a/4$, and its core has K coefficients at least a . The same bound applies. All coefficients multiplying the integrals are nonnegative. This also proves the upper bound by monotonicity. $\square$

C Resource-constrained path approximation

Proof of Lemma 6.2. Compute the minimum resource use $\ell_s(u)$ from the source to each vertex, and $\ell_t(v)$ from each vertex to the sink. If the minimum full-path resource exceeds B_H , the branch is infeasible. Otherwise an edge $e = (u, v)$ occurs on some feasible full path exactly when

$$\ell_s(u) + d_e + \ell_t(v) \leq B_H. \quad (\text{C.1})$$

Necessity is immediate. For sufficiency concatenate minimum-resource prefix and suffix paths with e . They cannot repeat a vertex in an acyclic graph. Delete all other edges. Every originally feasible path remains; the witness path for each retained edge remains too, since each of its edges belongs to that feasible path.

Let $P = \max_e b_e$ among retained edges. Nonnegative profits and a feasible path containing an edge of profit P imply

$$P \leq G^* \leq NP. \quad (\text{C.2})$$

If $P = 0$, return any feasible path. Otherwise set $\tau = \eta P/N$ and $\hat{b}_e = \lfloor b_e/\tau \rfloor$. Each $\hat{b}_e \leq N/\eta$, so a path's rounded profit is at most $\lfloor N^2/\eta \rfloor$. For every vertex v and rounded profit z , maintain

$D(v, z) =$ minimum *original* rational resource use to reach v with profit label z .

Initialize $D(\text{source}, 0) = 0$ and all other entries to $+\infty$. In topological order apply

$$D(v, z + \hat{b}_e) \leftarrow \min\{D(v, z + \hat{b}_e), D(u, z) + d_e\}.$$

Retaining the smallest resource use for a label loses no feasible continuation. Choose the largest terminal label whose stored resource is at most B_H . If π^* is an optimal feasible path and $\hat{\pi}$ is returned, then

$$\begin{aligned} G(\hat{\pi}) &\geq \tau \sum_{e \in \hat{\pi}} \hat{b}_e \geq \tau \sum_{e \in \pi^*} \hat{b}_e \geq G^* - N\tau \\ &= G^* - \eta P \geq (1 - \eta)G^*. \end{aligned}$$

Feasibility follows from exact resource sums and comparisons. There are $O(N^2/\eta)$ labels per vertex and the stated number of updates. All profit and resource inputs are rational with polynomial bit length, as are their path sums and the profit-scaling operations. $\square$

D Integral evaluation and arithmetic complexity

D.1 Relative integral approximation

Proof of Lemma 5.1. Relabel the coefficients so that $c_1 \geq c_2 \geq \dots \geq c_m > 0$; their order does not affect the integral. Let $S = s + \sum_\ell c_\ell$. Since $1 + x \leq e^x$, $f(t) \geq e^{-St}$ and $L \geq 1/S$. Also $f(t) \leq 1$ and $f(t) \leq (c_1 c_2 t^2)^{-1}$ for $t > 0$. Set

$$\ell = \frac{\delta}{8S}, \quad U_{\text{alg}} = \frac{8S}{\delta c_1 c_2}, \quad \gamma = 1 + \frac{\delta}{8}. \quad (\text{D.1})$$

For $s > 0$, choose the least nonnegative integer κ such that $2^\kappa \geq 8S/(\delta s)$ and put

$$U_{\text{exp}} = \frac{\kappa}{s}, \quad U = \min\{U_{\text{alg}}, U_{\text{exp}}\}.$$

For $s = 0$, put $U = U_{\text{alg}}$. All cutoffs are rational, and κ can be found by integer comparisons without evaluating a logarithm. The integrals over $[0, \ell]$ and $[U_{\text{alg}}, \infty)$ are each at most $\delta/(8S)$. When $s > 0$, the exponential tail gives

$$\int_{U_{\text{exp}}}^{\infty} f(t) dt \leq e^{-\kappa}/s \leq 2^{-\kappa}/s \leq \delta/(8S).$$

Thus the smaller cutoff U has the same tail guarantee. Moreover $U > \ell$: $S^2 \geq 4c_1 c_2$, $S \geq s$, and $\kappa \geq 1$. Form the rational geometric mesh $x_0 = \ell$, $x_{j+1} = \min\{\gamma x_j, U\}$ until $x_J = U$. Because f decreases, its right-endpoint sum Q is a lower bound on L . For every $t \in [x_j, x_{j+1}]$, $x_{j+1} \leq \gamma t$, so

$$\begin{aligned} Q &= \sum_{j=0}^{J-1} (x_{j+1} - x_j) f(x_{j+1}) \\ &\geq \int_{\ell}^U f(\gamma t) dt = \gamma^{-1} \int_{\gamma \ell}^{\gamma U} f(u) du \\ &\geq \gamma^{-1} L - \ell - \frac{\delta}{8\gamma S} \geq \left(1 - \frac{3\delta}{8}\right) L. \end{aligned}$$

This integrated comparison is essential: no pointwise relative control of $e^{-s\gamma t}/e^{-st}$ is needed.

Set $\nu = \delta/(16SU)$. At a node y , compute a rational E_y with $0 \leq e^{-sy} - E_y \leq \nu$. Divide it by the exact rational product $\prod_\ell (1 + c_\ell y)$, then round down to a dyadic grid of mesh $2^{-b} \leq \nu$. The resulting $\underline{f}(y)$ lies in $[0, f(y)]$ and has absolute error at most 2ν . The sum using $\underline{f}$ instead of f loses at most $2\nu U = \delta/(8S) \leq \delta L/8$. Thus

$$(1 - \delta/2)L \leq \underline{L} := \sum_j (x_{j+1} - x_j) \underline{f}(x_{j+1}) \leq L,$$

which implies the asserted, slightly weaker guarantee.

For completeness, the exponential lower approximation also uses rational arithmetic only. Choose an integer D with $2^{-D} \leq \nu$. If $x = sy \geq D$, take $E_y = 0$, since $e^{-x} \leq 2^{-D}$. Otherwise $0 \leq x < D$. Let $P_n = \sum_{j=0}^n x^j/j!$ and $v_{n+1} = x^{n+1}/(n+1)!$. Choose n so that $x/(n+2) \leq 1/2$ and $2v_{n+1} \leq \nu$. The positive Taylor tail is at most $2v_{n+1}$, hence

$$P_n \leq e^x \leq P_n + 2v_{n+1}, \quad E_y = (P_n + 2v_{n+1})^{-1} \leq e^{-x}, \quad e^{-x} - E_y \leq 2v_{n+1} \leq \nu.$$

The last estimate uses $P_n \geq 1$. A factorial bound shows that $n = O(D + \log(1/\nu))$ suffices.

If the rational input has total bit length b_0 , then $\log(U/\ell)$ and $\log(1/\nu)$ are polynomial in $b_0 + \log(1/\delta)$. The number of cells is $O(\delta^{-1} \log(U/\ell) + 1)$; mesh points, Taylor terms, dyadic approximations, and the final rational sum all have polynomial bit length. Even multiplication of all denominators gives a polynomial bound, since there are polynomially many terms of polynomial length. All comparisons, floors, products, and sums can therefore be performed in polynomial bit time. $\square$

D.2 Arithmetic complexity of the assortment algorithms

Under the input convention of Section 3, the running time of both algorithms is

$$N^{O(1/\epsilon)} \text{poly}(\mathcal{L}, 1/\epsilon)$$

in bit operations. Here $\mathcal{L}$ is the encoding length defined in Appendix A.3. To verify this bound, consider a branch with $K \leq N$. Each $w_i = q^{i-1}$ has bit length $O(N\mathcal{L})$. The grid width, rounded weights, integral shifts, and exact revenue multipliers have polynomial length as well. Lemma 5.1 therefore computes each edge score in polynomial time and returns a rational number of polynomial length.

Every path score is a sum of at most N edge scores. Such sums and their comparisons have polynomial bit complexity. For the constrained extension, resource sums, profit-scaling operations, and rounded-profit labels have polynomial length. Integral values are computed once per edge and reused across labels. The exact revenue evaluation in Appendix A.3 also takes polynomial time per candidate. Combining these costs with the numbers of states and updates in Sections 5.4 and 6 proves the bound.

E Certified branch bounds and accuracy tradeoffs

The head enumeration remains the main worst-case bottleneck. The following optional bounds permit safe pruning while preserving the approximation theorem. They also make explicit the scaling needed to compare a branch surrogate with an exact true-revenue incumbent.

E.1 Capacity relaxation and Lagrangian bounds

Fix H, Q and keep all count and outside-option restrictions in the graph. Let $U_{H,Q}$ be the maximum unscaled rational lower score, including $b_H(Q)$, over all source–sink paths after relaxing physical capacity. From (5.3), every feasible completion satisfies

$$R(H \cup T) \leq \frac{c_q}{\rho} F_H(T) \leq \frac{c_q}{\rho} U_{H,Q}. \quad (\text{E.1})$$

Thus maximizing a pointwise lower surrogate and dividing by its guaranteed relative factor gives an upper bound on true branch revenue.

When capacity is imposed, every rational $\lambda \geq 0$ yields the alternative unscaled bound

$$U_{H,Q}(\lambda) = b_H(Q) + \lambda B_H + \max_{\pi} \sum_{e \in \pi} (b_e - \lambda d_e), \quad (\text{E.2})$$

where the maximization drops physical capacity but retains the graph’s other restrictions. For a feasible path, adding $\lambda(B_H - \sum_e d_e) \geq 0$ to its score proves that (E.2) bounds its unscaled lower score from above. Signed edge rewards cause no difficulty for a longest-path DP on a DAG. Taking the minimum over any finite set of nonnegative rational multipliers, including zero, preserves validity and cannot weaken the zero-multiplier bound. No claim that a positive multiplier always improves it is required.

E.2 A rational comparison with true revenue

The constant c_q is generally irrational. For a safe rational upper bound, let

$$I_0 = \int_0^\infty \frac{dt}{(1+t)(1+qt)} = \frac{1}{c_q}.$$

This follows directly from Proposition 4.1 for $A = \{1\}$. For any rational $0 < \delta_c < 1$, evaluate it once with Lemma 5.1, obtaining $(1 - \delta_c)I_0 \leq \underline{I}_0 \leq I_0$. Then

$$\frac{1 - \delta_c}{\underline{I}_0} \leq c_q \leq c_q^+ := \min \left\{ 1, \frac{1}{\underline{I}_0} \right\}. \quad (\text{E.3})$$

The additional bound $c_q \leq 1$ follows from $-\log q \geq 1 - q$. At $q = 1$, simply use $c_q^+ = 1$. Both entries in the minimum are valid upper bounds, so the minimum is safe.

Consequently

$$\text{UB}_{H,Q} = \frac{c_q^+}{\rho} U_{H,Q}, \quad \text{UB}_{H,Q}(\lambda) = \frac{c_q^+}{\rho} U_{H,Q}(\lambda) \quad (\text{E.4})$$

are rational upper bounds on the true feasible branch optimum. A branch can be discarded if one such bound is at most the exact true revenue of an already feasible incumbent. If the optimal branch is discarded, the incumbent already matches or exceeds its best revenue. Pruning therefore preserves both Theorem 3.1 and Theorem 6.1. The accuracy δ_c affects bound tightness only and consumes none of the revenue-approximation budget. One may also use the simpler, weaker upper bound $c_q^+ = 1$ without the extra oracle call. The head contribution must be included in every bound, and the factor c_q must not be silently dropped when comparing with a true-revenue incumbent.

E.3 Explicit accuracy for a chosen head size

For any implementation choice $K \geq 4$, numerical relative tolerance $\delta_{\text{num}} \in (0, 1)$ and path tolerance $\delta_{\text{path}} \in [0, 1)$, the same argument yields the explicit factor

$$\left(1 - \frac{1}{4(K-3)} \right) \left(1 - \frac{1}{4(K-2)} \right) (1 - \delta_{\text{num}})(1 - \delta_{\text{path}}).$$

Here $\delta_{\text{path}} = 0$ when the branch is solved by exact longest path. All smaller assortments are evaluated exactly. This formula states an accuracy–runtime tradeoff; the prescribed K in (4.6) gives the PTAS. The complete algorithm is not claimed to be an EPTAS or an FPTAS.